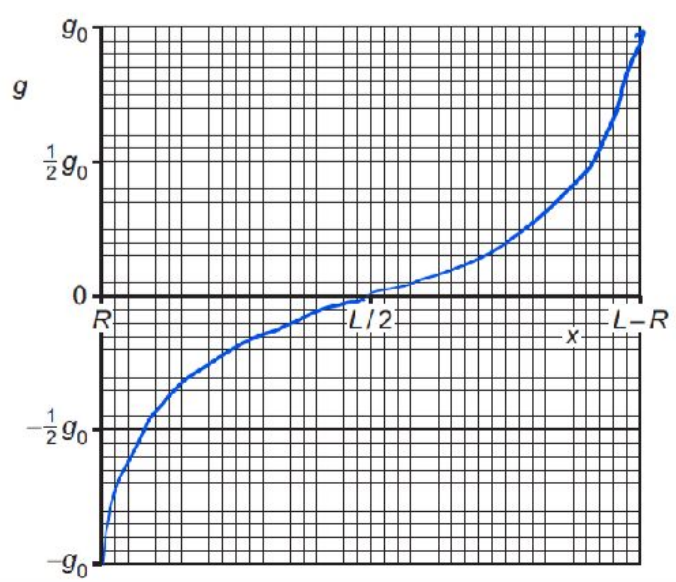
3(a)	Gravitational force provides the centripetal force. $G\frac{Mm}{r^2} = m\frac{v^2}{r}$ $G\frac{Mm}{r} = mv^2$ $\frac{1}{2}mv^2 = \frac{1}{2}\times G\frac{Mm}{r}$ Since $E_P = -G\frac{Mm}{r}$, $E_K = -\frac{E_P}{2}$	[1] [1]
(b)	From graph, $\phi = -3.2\times10^8$ J kg^{-1} at $r = 4.0\times10^8$ m. $E_{tot} = E_K + E_P = -\frac{E_P}{2} + E_P = \frac{E_P}{2} = \frac{m\phi}{2}$ $= \frac{8.93\times10^{22}\times(-3.2\times10^8)}{2}$ $= -1.43\times10^{31}$ J (negative sign essential) or 1. Use data from graph to find mass M of Jupiter using Ep = - GMm/r 2. Use E_T formula E_T = -½ GMm/r 3. Ans = -1.47 x 10^{31} or a bit off	[1] [1, ecf] [1]
(c)	Line starting at (R, $-g_0$) and ending at ($L - R$, $+g_0$) Line passing through (L/2, 0) Curve becomes less steep fr R to (L/2) and then steeper fr (L/2) to ($L - R$)  	[1] [1] [1]